AS YOU SIT DOWN

Does your setup work?

RUN THIS

Cmd/Ctrl + Shift + P → Create:
New Jupyter Notebook

```
import pandas as pd
import statsmodels.api as sm
print("pandas", pd.__version__)
print("statsmodels", sm.__version__)
```

Shift + Enter runs the cell. Two version numbers and you're set.

IF IT CAN'T FIND PANDAS

- You're on the wrong kernel — system Python, not Anaconda's
- The most common setup problem

Select Kernel (top right)
→ Python Environments
→ base or anaconda3

APPLIED ECONOMETRICS · RICE UNIVERSITY

Lab One

Setup, a course folder, and how AI is used in this course.

Jessica Rood
Teaching Assistant

jr120@rice.edu
Questions, any time

Sewall 207
Mondays

GOALS

What these labs are for

FIRST HALF · A WORKING SETUP

- Get your machine working, and keep it working
- Build a **workflow with AI** you can explain and defend
- Work through real data analysis, week by week

SECOND HALF · SOMETHING TO SHOW

- One guided project, start to finish
- In a **public repository on GitHub**
- Something you can hand an employer or an RA supervisor in December

Tonight is the first step of the first half.

TO START

How are you using AI right now?

WHICH COMPANY'S TOOL?

- GitHub Copilot
- Claude
- ChatGPT
- Gemini, Cursor, something else
- Nothing yet

IN A BROWSER, OR AGENTICALLY?

- A tab you paste into and read back from
- Or something that works **inside your project** — opens your files, runs your code

BEFORE ANY OF THE SETUP

AI in empirical research

WHERE THIS SITS

- Already in use: writing and debugging code, reorganising projects, catching errors
- **Not** for deciding what your results mean — that's the line the syllabus draws
- The distinction that matters: runs your code, or you paste into it

IF YOU WANT TO READ FURTHER

- **Scott Cunningham**
  Baylor; author of *Causal Inference: The Mixtape*
- Writing since Dec 2025 on how these tools are actually used in economics
- `causalinf.substack.com`
  `Optional, and partly paywalled`

THE DIFFERENCE THAT MATTERS

Can it see your files?

INSIDE YOUR EDITOR

- The file you have open
- The error it just produced
- The folder around it

IN A BROWSER TAB

- Only what you paste into it

Often the same model underneath. What differs is what it can see.

FROM THE SETUP GUIDE

What you need for today

1 · VSCODE

- code.visualstudio.com
- macOS: drag it into Applications
  Don't run it from Downloads
- Windows: tick **Add to PATH**

2 · ANACONDA

- anaconda.com/download
- Free account
- Brings pandas, numpy, statsmodels and matplotlib with it

3 · EXTENSIONS

- **Python**
  Microsoft
- **Jupyter**
  Microsoft
- An **AI extension**
  Your choice — this week, not tonight

Python and Jupyter is what the guide asked for. Missing one? Don't start downloading now — follow along and install afterwards.

GETTING ONE INTO VSCODE

Installing an extension

THE MECHANISM

Cmd/Ctrl + Shift + X

→ search the name

→ Install

→ sign in when it asks

The same four steps whichever tool you pick.

WHERE YOU ARE

Three answers

GREEN

- Two version numbers
- Help the person next to you

YELLOW

- An error — usually the kernel
- Your buddy or I will get to it

RED

- Not installed yet
- Don't start the download now — too large for the room's wifi
- Follow along; install afterwards

BEFORE WE BUILD ANYTHING

Two files, two jobs

README.MD

- What this is, and how you work
- You write the part about how you use AI
- Your assistant reads the rest every time it opens the folder

DECISIONLOG.MD

- What you did, and why
- The assistant writes it as you work
- You read what it wrote

Both sit at the top of `ECON209/`, covering homework and labs both.

NOW WATCH

Let it plan the whole thing

ASK FOR A PLAN FIRST

```
Set up my ECON209 folder:
homework/ and labs/, each with
data, code and output. Add
labs/project/ for the public
repo later. Draft README.md and
decisionlog.md at the top.
Plan it before you make anything.
```

- Folders **and** both files, in one pass
- It proposes the whole thing, then waits

THEN READ THE PLAN

- Does it match what you asked for?
- Is anything in the wrong place?
- Approve it, change it, or say no

The same lesson as reading a diff, one level up: the review happens **before** anything gets written.

WHAT THE PLAN PRODUCED

WHAT THE PLAN PRODUCED

Your course folder

```
ECON209/
├── README.md
├── decisionlog.md
├── homework/
│   ├── data/
│   ├── code/
│   └── output/
└── labs/
    ├── data/
    ├── code/
    ├── output/
    └── project/
```

THE RULES

- **data/** raw, never edited by hand
- **code/** your .ipynb notebooks
- **output/** made by code, safe to delete
- Both files at the top cover the whole course
- **project/** is the only thing that goes public — from Lab 6, with its own README
- Documents, not Downloads

WHAT IT DRAFTED, AND THE PART IT SHOULD NOT

README.md

```
# ECON 209 — your name

## How I use AI here
The syllabus allows AI for coding on problem
sets — not for interpreting results, not for
non-coding questions, never on exams.
I use it for ... I don't use it for ...
                        ← this part is yours

## How this folder works
homework/  problem sets, stays private
labs/      weekly lab work
both hold data/, code/ and output/
labs/project/  the public repo, from Lab 6

## Where things stand
Lab 1, 24 Aug — environment working,
folders built. No analysis yet.
```

READ IT, THEN TAKE ONE PART BACK

- It drafted this — check it against what you asked for
- **Rewrite the middle paragraph yourself** — how you use AI is the one thing it shouldn't write
- It keeps the last section current
- Private. **labs/project/** gets a separate README later, for the public

DRAFTED ALONGSIDE IT

decisionlog.md

```
# Decision log

## 2026-08-24 · Lab 1
- Set up VSCode against the Anaconda kernel.
- Built homework/ and labs/, each with
  data, code and output.
- Picked GitHub Copilot — it was the one
  that ran inside the editor for free.
- Kept the sample data untouched; wrote a
  copy into output/ instead.
```

STARTED FOR YOU

- The first entry is the folder it just built — **read what it wrote**
- Date, what you did, and **why**
- The why is what you'll want in November, when you can't remember which specification you settled on

THE POLICY, IN PRACTICE

Where the line is

FINE

+ Check my code for bugs
+ What does this error mean?
+ How do I get robust standard errors?
+ Write a loop that runs this for each state

NOT FINE

- Check my interpretation paragraph
- What does this coefficient mean?
- Is this effect causal?
- Anything, in any form, on an exam

THE STANDARD FOR THIS COURSE

You have to be able to explain any number you report

Where the estimate came from, what it assumes, and why you find it credible. Unauthorized use of AI is an Honor Code violation, and you sign this on every problem set.

WRAPPING UP

Before Lab 2

DO THIS WEEK

- Create a **GitHub account**
- Apply for the **Student Developer Pack**
  education.github.com/pack

BRING

- A working Python setup
- Tonight's folder

NEXT WEEK

- Real data — the Card and Krueger minimum wage study

Pack approval can take about three days. Apply from campus, not on a VPN, with dated proof of enrolment matching your GitHub name — a dated class schedule works better than a student ID. `jr120@rice.edu`